
Multivariate bandit pricing: why it is immune from the curse of dimensionality

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Abstract

1 The construction of [Castiglioni et al. \[2026\]](#) implies that

2 1 Introduction

3 Dynamic pricing is one of the earliest and most natural application domains for bandit algorithms. In
4 its simplest form, the problem is defined by a compact set of prices $\mathcal{X} \subset [0, 1]$ and a demand function
5 $D : \mathcal{X} \rightarrow [0, 1]$, which represents the probability of completing a transaction at a given price. An
6 agent interacts with the environment with the goal of maximizing the revenue function

$$J(p) = p \cdot D(p),$$

7 which captures the trade-off between the revenue obtained upon purchase, equal to p , and the
8 probability of completing the transaction, which typically decreases with price. The main challenge
9 is that the demand function D is unknown to the agent. Instead, the only way to gather information
10 about it is to experiment with different prices and observe the outcomes of transactions.

11 This structure makes dynamic pricing a natural instance of a bandit problem, an observation that
12 dates back to early work in economics [Rothschild \[1974\]](#) and was later formalized in the machine
13 learning literature by [Kleinberg and Leighton \[2003\]](#). In particular, the problem can be viewed as
14 learning a demand curve from binary feedback: at each round t , the agent selects a price p_t and
15 observes a stochastic outcome indicating whether the sale occurs or not. In this setting, the natural
16 way to measure the performance of an agent is through its cumulative revenue over T rounds, i.e.,
17 $\sum_{t=1}^T J(p_t)$. Since the demand is stochastic, the quantity $J(p)$ should be interpreted as the expected
18 revenue at price p . Therefore, performance is typically evaluated in expectation. Maximizing
19 cumulative revenue is equivalent to minimizing the gap with respect to the optimal fixed price in
20 hindsight, defined as $J^* := \sup_{p \in \mathcal{X}} J(p)$. This leads to the notion of (pseudo-)regret, defined as

$$R_T = \sum_{t=1}^T (J^* - J(p_t)).$$

21 The goal in this line of work is to design algorithms whose regret grows sublinearly in T , so that
22 the average regret R_T/T vanishes as $T \rightarrow \infty$. As we discuss in section 6, several works have
23 studied this problem. Building on the seminal formulation of [Kleinberg and Leighton \[2003\]](#), most
24 of the literature focuses on the one-dimensional setting and achieves regret guarantees of order
25 $R_T = \tilde{O}(T^{2/3})$, under the assumption that the demand curve corresponds to the survival function
26 of a random valuation. A natural extension of this model is the multivariate setting, in which
27 multiple products are priced simultaneously. Despite its practical relevance, this setting remains
28 poorly understood from a theoretical perspective. Indeed, moving from one to multiple dimensions
29 fundamentally alters the statistical structure of the problem, making it significantly more challenging
30 to obtain nonparametric regret guarantees.

31 1.1 Hardness of multi-variate pricing

32 In the generalized pricing problem, the agent selects at each round a vector of prices $\mathbf{p}_t \in \mathcal{X} \subset [0, 1]^d$,
 33 one for each product. The main challenge in this setting stems from the interaction between products,
 34 which requires learning a multivariate demand function. Without additional structure, this problem
 35 reduces to the general framework of continuum-armed bandits. Under assumptions analogous to
 36 those of [Kleinberg and Leighton \[2003\]](#), this leads to regret rates of order

$$R_T = \tilde{\Theta}\left(T^{\frac{d+1}{d+2}}\right),$$

37 implying that the average regret decays as $T^{-\frac{1}{d+2}}$. This rate deteriorates rapidly with the dimension
 38 d , highlighting a fundamental curse of dimensionality [Agrawal \[1995\]](#), [Kleinberg \[2004\]](#). Several
 39 approaches have been proposed to mitigate this issue, but all of them present significant limitations,
 40 as we substantiate in section 6.

41 A recent attempt to extend the original nonparametric framework of [Kleinberg and Leighton \[2003\]](#)
 42 to the multivariate setting is [Castiglioni et al. \[2026\]](#). In their model, d products are priced simulta-
 43 neously, and the feedback is of the same one-bit nature as in the univariate case: at each round, the
 44 agent only observes whether a transaction occurs or not.

45 More precisely, the buyer is characterized by a random valuation vector $\mathbf{A}_t \in [0, 1]^d$, and accepts
 46 the offer if and only if the proposed price vector satisfies $\mathbf{p}_t \leq \mathbf{A}_t$ component-wise. Under this
 47 assumption, the demand function takes the form

$$\mathbf{D}(\mathbf{p}) = \mathbb{P}(\mathbf{p} \leq \mathbf{A}), \quad \mathbf{v} \leq \mathbf{u} \iff \forall i = 1, \dots, d, v_i \leq u_i,$$

48 which is the natural multivariate analogue of the survival function considered in [Kleinberg and](#)
 49 [Leighton \[2003\]](#). The corresponding revenue function is

$$J(\mathbf{p}) = \left(\sum_{i=1}^d p_i \right) \mathbf{D}(\mathbf{p}).$$

50 Remarkably, this simple structural assumption allows [Castiglioni et al. \[2026\]](#) to design an algorithm
 51 achieving regret

$$R_T = \tilde{O}\left(T^{3/4}\right),$$

52 independently of the dimension d . This result stands in stark contrast with the curse of dimensionality
 53 exhibited by general nonparametric multivariate pricing models. The goal of this paper is to further
 54 investigate this regime, extending these guarantees and understanding the fundamental limitations of
 55 this approach.

56 1.2 Our Contribution

57 The structural assumption of [Castiglioni et al. \[2026\]](#) implies that the demand function is monotone
 58 with respect to prices, as it is given by the survival function of a multivariate valuation distribution.
 59 However, this property is known to be violated in many realistic settings, including entire classes of
 60 goods such as luxury products, where demand may increase with price; we discuss this in detail in
 61 section 6. In this work, we show that this assumption can be removed by introducing a more general
 62 model of consumer behavior. In addition to a maximal valuation vector $\mathbf{A}_t$, we consider a lower
 63 valuation vector $\mathbf{B}_t$, representing the fact that a user may also refuse to purchase if prices are too low.
 64 Under this model, a transaction occurs if and only if $\mathbf{B}_t \leq \mathbf{p}_t \leq \mathbf{A}_t$ component-wise.

65 This additional layer of complexity fundamentally alters the structure of the problem and requires
 66 a different algorithmic approach. While [Castiglioni et al. \[2026\]](#) relies on an explore-then-commit
 67 strategy, we develop a method based on embedding the class of objective functions into a Sobolev
 68 space with dominating mixed smoothness. This allows us to reduce the problem to a special kind of
 69 kernelized bandit setting. Under suitable integrability assumptions on the marginal distributions of
 70 $(\mathbf{B}_t, \mathbf{A}_t)$, we obtain a regret bound of order

$$R_T = \tilde{O}\left(T^{3/4}\right),$$

71 matching the best known guarantees while significantly relaxing the structural assumptions; see
 72 theorem 4.3. Moreover, we show that this rate is optimal under our assumptions, as established in
 73 section 4.2.

74 To the best of our knowledge, this is the first work that combines multivariate dynamic pricing with
 75 tools from Sobolev spaces and kernelized bandits. Beyond pricing, our analysis opens the door to a
 76 broader class of bandit problems involving structured high-dimensional objectives.

77 2 Multivariate Pricing Model

78 Denote $d \in \mathbb{N}$ the dimension of our multivariate pricing model, which corresponds to the number of
 79 products we take into consideration. In this work, we consider a pricing problem on discrete time
 80 steps $t = 1, \dots, T$, where $T \in \mathbb{N}$ is the time horizon. At any step, the interaction starts with the agent
 81 posting a vector of prices $\mathbf{p}_t \in [0, 1]^d$. Then, a customer arrives characterized by two valuations $\mathbf{A}^{(t)}$
 82 and $\mathbf{B}^{(t)}$, corresponding to random vectors of size d , equal to the number of products. The customer
 83 buys at the proposed price vector $\mathbf{p}_t$ if $\mathbf{B}_t \leq \mathbf{p}_t \leq \mathbf{A}_t$ in all the dimension simultaneously, meaning
 84 that the price of each product i lies in the range $B_{t,i}, A_{t,i}$ that satisfies the customer.

As the different product may be related one to each other, we assume that, for some probability
 distribution $\mathcal{P}$,

$$\mathbf{B}_t, \mathbf{A}_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{P}([0, 1]^{2d}).$$

85 This correspond to the fact that different clients come with i.i.d. valuations but the two vectors $\mathbf{B}_t, \mathbf{A}_t$
 86 can be correlated in any possible way: both $A_{t,i}$ with $A_{t,j}$ (upper valuation for different products)
 87 and $B_{t,i}$ with $A_{t,i}$ (lower and upper valuation for the same product). In this model, the demand and
 88 objective functions write as follow.

Definition 2.1 (Objective function). The demand D and objective J functions, both $[0, 1]^d \rightarrow \mathbb{R}_+$,
 write as

$$D(\mathbf{p}) := \mathbb{P}(\mathbf{B}_1 \leq \mathbf{p} \leq \mathbf{A}_1) \quad J(\mathbf{p}) := D(\mathbf{p}) \sum_{i=1}^d p_i.$$

89 Interestingly, as similar model applies to many framework that go much beyond pure pricing, as we
 90 discuss in section 6.1. The performance metric used to quantify the behavior of the agent, measuring
 91 how much it differs from a clairvoyant that always plays the maximum of $J(\mathbf{p})$, is the pseudo-regret.

Definition 2.2 (Pseudo-Regret). The cumulative pseudo-regret R_T is defined as:

$$R_T = \sum_{t=1}^T \left[\sup_{\mathbf{p} \in [0, 1]^d} J(\mathbf{p}) - J(\mathbf{p}_t) \right].$$

92 If R_T grows sub-linearly with T , it means that $J(\mathbf{p}_t)$ is, on average, converging to the best possible
 93 performance. Finding a policy whose regret is as small as possible is the goal of theoretical studies
 94 on online pricing. As we show in section 4.2 this problem does not admit solutions without further
 95 assumptions, as one could have Dirac-type distributions for $\mathbf{A}_1, \mathbf{B}_1$ that are arbitrarily close one to
 96 each other. This "needle in a haystack" may be solved with the following assumption, that ensure that
 97 neither $\mathbf{B}_1$ nor $\mathbf{A}_1$ are too concentrated.

Assumption 2.3. The two random vectors $\mathbf{B}, \mathbf{A}$ admit marginals probability density functions f_B
 and f_A given, respectively, that satisfy, for some $L_2 > 0$,

$$\int_{[0, 1]^d} f_B(b)^2 db \leq L_2, \quad \int_{[0, 1]^d} f_A(a)^2 da \leq L_2.$$

98 This assumption includes the vast majority of common-use probability distributions on $[0, 1]^d$ and,
 99 importantly, do not imply boundedness. While any bounded probability density function satisfies
 100 assumption 2.3, distribution like Beta(α, β) for $1/2 < \alpha, \beta < 1$ satisfy only assumption 2.3.

3 Reduction to Dominating Mixed Smoothness Spaces

The main technical contribution of this work is to show that the multivariate pricing problem introduced in Section 2 admits a hidden regularity structure, which allows us to bypass the curse of dimensionality. At first glance, the objective function $J(\mathbf{p}) = D(\mathbf{p}) \sum_{i=1}^d p_i$ may appear a generic function, difficult to characterize. Indeed, the demand function $D(\mathbf{p}) = \mathbb{P}(\mathbf{B} \leq \mathbf{p} \leq \mathbf{A})$ is defined through a probability of a multidimensional orthant event, and does not satisfy standard assumptions such as monotonicity or classical smoothness. As a consequence, existing approaches based on Lipschitz or Hölder regularity would lead to regret bounds suffering from an exponential dependence on the dimension d .

Our key observation is that the structure of the event $\{\mathbf{B} \leq \mathbf{p} \leq \mathbf{A}\}$ induces a much more favorable form of regularity. In particular, although D may fail to be smooth in the classical sense, its *mixed partial derivatives* exhibit a simple probabilistic structure. Differentiating with respect to a coordinate p_i does not create arbitrary singularities, but instead exposes marginal densities of the random vectors $\mathbf{A}$ and $\mathbf{B}$. As a result, the complexity of D is not governed by isotropic smoothness (which would depend exponentially on d), but by a weaker and anisotropic notion known as *dominating mixed smoothness*. This notion captures functions whose mixed derivatives are square-integrable, and is known to admit approximation rates that depend only mildly on the dimension. We now formally introduce the function space that will be used throughout the paper.

Definition 3.1 (Dominating mixed smoothness of order 1). The Hilbert space $H_{\text{mix}}^1(\mathcal{X})$ of functions with dominating mixed smoothness on $\mathcal{X}$ is the set of functions such that for every multi-index $\alpha \in \{0, 1\}^d$, the mixed weak derivative $\partial^\alpha f$ exists and belongs to $L^2(\mathcal{X})$. The norm in this space writes as follows

$$\|f\|_{H_{\text{mix}}^1}^2 := \sum_{\alpha \in \{0,1\}^d} \|\partial^\alpha f\|_{L^2}^2.$$

Standard isotropic Sobolev spaces are defined by summing over multi-indices α such that $|\alpha| \leq 1$, where $|\alpha| = \sum_{i=1}^d \alpha_i$. In this case, only first-order partial derivatives along individual coordinates are taken into account. In contrast, definition 3.1 involves a sum over all $\alpha \in \{0, 1\}^d$, which includes mixed derivatives of all orders, up to the full derivative $\alpha = (1, 1, \dots, 1)$. As a result, this space captures functions whose complexity is governed by the interaction between coordinates, rather than by their individual smoothness alone.

We are now ready to state the main structural result of the paper. The following theorem shows that the probabilistic model introduced in Section 2 induces a strong form of regularity on the objective function, *despite the absence of classical isotropic smoothness*, which would require unrealistic assumptions on the derivative of the valuations' densities.

Theorem 3.2. Under assumption 2.3, the objective function 2.1 belongs to the space $H_{\text{mix}}^1([0, 1]^d)$ introduced in definition 3.1, and $\|J\|_{H_{\text{mix}}^1}^2 \leq d^2 2^{d+4} L_2$.

Proof sketch. The key idea is that mixed derivatives of the objective function $J(\mathbf{p}) = D(\mathbf{p}) \sum_{i=1}^d p_i$ inherit a simple structure from the demand function $D(\mathbf{p}) = \mathbb{P}(\mathbf{B} \leq \mathbf{p} \leq \mathbf{A})$. We first focus on D . By expressing it as

$$D(\mathbf{p}) = \int_{[0, \mathbf{p}]} \int_{[\mathbf{p}, 1]} f_{B,A}(b, a) \, da \, db,$$

one sees that each coordinate p_i appears as a boundary of the integration domain. As a consequence, differentiating with respect to p_i amounts to exposing boundary terms involving the joint density $f_{B,A}$. In particular, taking a full mixed derivative $\partial^\alpha D$ with $\alpha \in \{0, 1\}^d$ yields a finite sum of terms where one argument of the density is fixed at $\mathbf{p}$ while the others are integrated out. These expressions can be controlled in L^2 by the marginal densities f_A and f_B . Under assumption 2.3, this implies that all mixed derivatives of D up to order d belong to $L^2([0, 1]^d)$, with a bound proportional to L_2 .

The result for J then follows by applying the product rule. Since $\sum_{i=1}^d p_i$ is a linear function, its derivatives are uniformly bounded, and each mixed derivative of J can be expressed as a combination of derivatives of D of order at most $|\alpha|$. This preserves the L^2 bounds and yields $\|J\|_{H_{\text{mix}}^1}^2 \leq d^2 2^{d+4} L_2$.

The full proof is given in Appendix B. Theorem 3.2 shows that our pricing problem admits a favorable functional structure: the objective function J belongs to a Sobolev space with dominating mixed smoothness $H_{\text{mix}}^1([0, 1]^d)$. This result is purely structural, and does not yet provide an algorithm. To turn this observation into a learning procedure, we need to connect this function space to a bandit framework with existing and known regret guarantees. Our approach is to embed $H_{\text{mix}}^1([0, 1]^d)$ into a reproducing kernel Hilbert space (RKHS), so that we can leverage the well-established theory of kernelized bandits. This is the topic for the next section.

This raises a crucial question: *which kernel should be used?* Ideally, we seek a kernel whose RKHS is as small as possible, while still containing $H_{\text{mix}}^1([0, 1]^d)$. This is essential to obtain tight control on the complexity of the problem, as overly large RKHSs would lead to loose regret bounds.

As we show below, a natural and essentially optimal choice is given by a tensorized Matérn kernel. This construction mirrors the anisotropic structure of dominating mixed smoothness, and leads to a tight embedding of $H_{\text{mix}}^1([0, 1]^d)$ into a kernel space with favorable information-theoretic properties.

3.1 Construction of equivalent kernel

Which kernel should be used? Ideally, we seek a kernel whose RKHS is as small as possible, while still containing $H_{\text{mix}}^1([0, 1]^d)$. This is essential to obtain tight control on the complexity of the problem, as overly large RKHSs would lead to loose regret bounds.

A standard and very well-studied choice for embedding Sobolev spaces into RKHS is the Matérn kernels Matérn [2013]. In fact, it is well-known that, in dimension $d = 1$, the Sobolev space of functions with weak derivative in L^2 is equivalent to the RKHS generated by the Matérn kernel with parameters $\nu = 1/2$, which takes the form

$$\forall x, y \in \mathbb{R} \quad k(x, y) := \exp(-|x - y|). \quad (1)$$

This kernel is Laplace’s covariance function. While generalizations of this kernel exist in arbitrary dimension d , the corresponding Hilbert spaces are equivalent to isotropic Sobolev spaces; which is exactly what we are trying to avoid. In particular, such kernels fail to capture the anisotropic structure highlighted in Section, where mixed derivatives govern the complexity of the problem. An examination of the spectrum of multidimensional Matérn kernels reveals that the eigenvalues corresponding to the eigenfunctions are larger than necessary for a function with dominating mixed smoothness, where the interaction between multiple variables is weak.

Rather than relying on existing multivariate kernels, we solve this issue by applying the Laplace kernel on each of the component, and making the product.

Definition 3.3 (Product kernel). The product kernel on $\mathbb{R}^d$ writes as

$$\forall x, y \in \mathbb{R}^d \quad k_{\Pi}(x, y) = \prod_{i=1}^d k(x_i, y_i),$$

where k comes from (2). We name the RKHS generated by k_{Π} on some set $\mathcal{X} \subseteq \mathbb{R}^d$ as $\mathcal{H}_{\Pi}^k(\mathcal{X})$

This seemingly straightforward technique is based on a profound intuition. When taking the product of a kernel—whose eigenvalues decay polynomially—with itself, the ‘mixed’ components corresponding to indices i, j that are both moderate result in an eigenvalue $\mu_i \mu_j$ that is exceedingly small.

The next theorem shows that this kernel provides the desired embedding.

Theorem 3.4. [Space equivalence] *There is a constant C_d only depending on d such that, for any $f \in H_{\text{mix}}^1([0, 1]^d)$ we have $f \in \mathcal{H}_{\Pi}^k([0, 1]^d)$ and*

$$\|f\|_{\mathcal{H}_{\Pi}^k} \leq C_d \|f\|_{H_{\text{mix}}^1}.$$

3.2 Construction of equivalent kernel

The Matérn kernel of parameters $d = 1, \nu = 1/2$ writes as follows

$$\forall x, y \in \mathbb{R} \quad k(x, y) := \exp(-|x - y|) \quad (2)$$

4 Algorithm

First, we reduce the problem to a finite number of arms.

Lemma 4.1. *Fix $\varepsilon > 0$. Then, for any ε -cover $\mathcal{X}_\varepsilon$ of $\mathcal{X}$,*

$$\max_{\mathbf{p} \in \mathcal{X}_\varepsilon} J(\mathbf{p}) \geq \sup_{\mathbf{p} \in \mathcal{X}} J(\mathbf{p}) - C_\rho \sqrt{L_2 \varepsilon},$$

for some constant C_ρ only depending on d .

Proof. By the immersion theorems (chapter 2 of [Schmeisser and Triebel \[1987\]](#)) any function $f \in H_{\text{mix}}^1([0, 1]^d)$ is $1/2$ -Hölder continuous with a constant $C_H \|f\|_{H_{\text{mix}}^1}$. Now, take $\mathbf{p}^* \in \mathcal{X}$ such that $\sup_{\mathbf{p} \in \mathcal{X}} J(\mathbf{p}) - J(\mathbf{p}^*) \leq \sqrt{\varepsilon}$. Then, there is an element $\mathbf{p}'$ in the cover such that $\|\mathbf{p}' - \mathbf{p}^*\|_\infty \leq \varepsilon$. Therefore, by Hölder continuity,

$$J(\mathbf{p}') \geq J(\mathbf{p}^*) + C_H \|f\|_{H_{\text{mix}}^1} \sqrt{\varepsilon} \geq \sup_{\mathbf{p} \in \mathcal{X}} J(\mathbf{p}) - (C_H \|f\|_{H_{\text{mix}}^1} + 1) \sqrt{\varepsilon}.$$

Replacing the function norm with theorem 3.2 and naming $C_\rho := C_H 2^{d/2+2} d$ ends the proof. $\square$

Importantly, a cover $\mathcal{X}_\varepsilon$ can be chose of size ε^{-d} . For $\varepsilon = 1/T$, the former result ensures that passing to this cover has a negligible effect on the regret. For this reason, we shall call

$$\mathcal{X}^\circ := \mathcal{X}_{1/T} \quad R_T^\circ = \sum_{t=1}^T J(x^\circ) - J(x_t), \quad (3)$$

knowing that $R_T - R_T^\circ \leq \dots$

We use BPE (algorithm 1 from [Li and Scarlett \[2022\]](#)) for some choice of the kernel

4.1 Bounding the information gain

Theorem 4.2. *The maximum information gain for kernel k_Π is bounded as follows:*

$$\gamma_T \leq C_\gamma \sqrt{T} \log(T)^{d+1/2}.$$

Theorem 4.3. *[Regret bound] With probability at least $1 - \delta$*

$$R_T \lesssim \left(L_2 + \sqrt{\log(T/\delta)} \right) \log(T)^{d+1/2} T^{3/4}.$$

Proof. Theorem 1 from [Li and Scarlett \[2022\]](#) guarantees a bound on the regret of the form

$$R_T \leq \mathcal{O} \left(\Lambda \sqrt{\gamma_T T} \right)$$

For $\Lambda = \|J\|_{\mathcal{H}_\Pi^k} + \sqrt{\log(|\mathcal{X}^\circ|/\delta)}$. We can upper-bound the two quantities separately:

1. By (3) and theorem 3.2 and theorem 3.4, the first term is bounded by

$$\Lambda = \|J\|_{\mathcal{H}_\Pi^k} + \sqrt{\log(|\mathcal{X}^\circ|/\delta)} \lesssim \sqrt{L_2} + \sqrt{\log(T/\delta)}.$$

2. By theorem 4.2 the information gain is bounded by $\gamma_T \leq C_\gamma \sqrt{T} \log(T)^{d+1/2}$.

This proves that $R_T^\circ \lesssim (L_2 + \sqrt{\log(T/\delta)}) \log(T)^{d+1/2} T^{3/4}$. $\square$

197 4.2 Lower Bound

198 Riusciamo a dimostrare che se assumption 2.3 è violata il regret cresce oltre $T^{3/4}$?

199 Controesempio: define, for $x_0 \in [0, 1]$ and $\beta \in (1/2, 1)$

$$f_A(x|x_0, \beta, \varepsilon) = \frac{1}{10} + \frac{9(1-\beta)\mathbb{I}\{x \in [x_0, x_0 + \varepsilon]\}}{10\varepsilon^{1-\beta}(x-x_0)^\beta}$$

$$f_B(x|x_0, \beta, \varepsilon) = \frac{1}{10} + \frac{9(1-\beta)\mathbb{I}\{x \in [x_0 - \varepsilon, x_0]\}}{10\varepsilon^{1-\beta}(x_0-x)^\beta}$$

200 Todo:

- 201 • verify that they integrate to one
- 202 • verify for which α we have $f_B^\alpha \in L^1$ (it will be a function of β)

203 4.3 Is case of bounded distribution easier?

204 5 Numerical Simulations

205 6 Related Works

206 **Classical Pricing** REFERENCE REVIEW [Mussi \[2026\]](#)

207 [Araman and Caldentey \[2009\]](#) use a Bayesian approach for dynamic pricing: here, market-related
208 information is captured in the model through a prior belief on parameters, and the model has a
209 monotonic formulation on the demand function.

210 [Bu et al. \[2020\]](#) nulla da dire (offline)

211 [Kleinberg and Leighton \[2003\]](#) fondativo

212 In [Misra et al. \[2019\]](#), the monotonic property of the demand function is used to guarantee a faster
213 convergence. However, monotonicity is not enforced as a model-specific feature, so decisions
214 violating business logic can still be made during the learning process.

215 [Trovo et al. \[2015\]](#), [Trovò et al. \[2018\]](#) assume that the demand function is monotonically decreasing
216 and exploit this assumption in the learning algorithm to provide uncertainty bounds tighter than those
217 of classical frequentist MAB algorithms, but neither monotonicity nor weak monotonicity is imposed
218 by the model formulation on the estimated demand functions. The authors show that the monotonicity
219 assumption does not improve the asymptotic regret bound provided by MAB theory. On the other
220 hand, exploiting monotonicity enables empirical improvements in performance, thereby reducing the
221 constant terms in the regret bound.

222 [Rothschild \[1974\]](#) (paper seminale antichissimo, per dire che bandit è il primo problema pricing)
223 provides one of the seminal works on the adoption of MAB algorithms for dynamic pricing. In this
224 work, the author introduces a basic scenario in which a two-armed bandit has to decide which price
225 is better between the two options. Using this work as a starting point, many others tried to face the
226 dynamic pricing problem using MABs.

227 [Wang et al. \[2021\]](#) investigate non-parametric models for demand function estimation. In this case,
228 the authors assume that the demand function is smooth.

229 [Mussi et al. \[2022\]](#) modern work that faces product aggregation with ts style algorithm.

230 **Multivariate pricing** A common strategy is to impose a parametric structure on the demand
231 function, as in [Perivier and Goyal \[2022\]](#), [Javanmard et al. \[2020, 2024\]](#). While this improves regret
232 guarantees, it comes at the cost of significantly restricting the expressiveness of the model.

233 An alternative direction is to retain a nonparametric formulation while introducing regularity as-
234 sumptions. For instance, [Yang et al. \[2024\]](#) obtains improved regret rates under strong concavity
235 assumptions on the revenue function. However, such assumptions can be difficult to justify in practice.

236 A more general approach is proposed by [Besbes and Zeevi \[2012\]](#), who consider demand functions
237 with s bounded derivatives and obtain regret rates of order

$$\tilde{\Theta}\left(T^{\frac{d+s}{d+2s}}\right),$$

238 without requiring concavity. Nevertheless, achieving substantial improvements over the baseline
239 continuum bandit rates would require s to scale with d , which is unrealistic in many practical scenarios
240 where demand functions exhibit non-smooth behavior, such as spikes or kinks.

241 Non monotonical demand functions

242 **Mixed smoothness** [Bungartz and Griebel \[2004\]](#)

243 6.1 Similar Models

244 The objective function that we defined in definition 2.1 is actually an archetype that applies for several
245 problems.

- 246 • Clinical trials
- 247 • Survivor of an electrical system
- 248 • Advertising

249 7 Conclusions

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312 A Notation

Symbol	Meaning
T	Time horizon.

313 B Missing proofs

Lemma B.1. *Let $D(\mathbf{p}) = \mathbb{P}(\mathbf{B} \leq \mathbf{p} \leq \mathbf{A})$. Under assumption 2.3, for every $\alpha \in \{0, 1\}^d$ one has*

$$\|\partial^\alpha D\|_{L^2}^2 \leq 4L_2.$$

314 *Proof.* By definition,

$$\begin{aligned} D(\mathbf{p}) &= \mathbb{P}(\mathbf{B} \leq \mathbf{p} \leq \mathbf{A}) \\ &= \int_{[0, \mathbf{p}]} \int_{[\mathbf{p}, \mathbf{1}]} f_{B,A}(b, a) \, da \, db. \end{aligned}$$

315 Note that $f_{B,A}(b, a)$ is not intended to be a proper function as this point, and is allowed to have
316 Dirac-like parts. We prove the statement by backward induction. Starting from $\alpha = (1, 1, \dots, 1)$, we
317 prove that for any multi-index such that $\sum_{i=1}^d \alpha_i = k$, $\partial^\alpha D \in L^2([0, 1]^d)$ with squared-norm not
318 exceeding $4L_2$.

For the base case, $\alpha = (1, 1, \dots, 1)$ and

$$\partial^\alpha D(\mathbf{p}) = \int_{[\mathbf{p}, \mathbf{1}]} f_{B,A}(\mathbf{p}, a) \, da - \int_{[0, \mathbf{p}]} f_{B,A}(b, \mathbf{p}) \, db.$$

319 Therefore,

$$\|\partial^\alpha D\|_{L^2}^2 \leq 2 \int_{[0, 1]^d} \left(\int_{[\mathbf{p}, \mathbf{1}]} f_{B,A}(\mathbf{p}, a) \, da \right)^2 d\mathbf{p} + 2 \int_{[0, 1]^d} \left(\int_{[0, \mathbf{p}]} f_{B,A}(b, \mathbf{p}) \, db \right)^2 d\mathbf{p}$$

$$\begin{aligned} &\leq 2 \int_{[0,1]^d} f_B(\mathbf{p})^2 d\mathbf{p} + 2 \int_{[0,1]^d} f_A(\mathbf{p})^2 d\mathbf{p} \\ &\leq 4L_2. \end{aligned}$$

320 The second passage comes from the fact that the integral of a positive function over $[\mathbf{0}, \mathbf{p}]$ or $[\mathbf{p}, \mathbf{1}]$ is
 321 upper bounded by its integral on the whole domain. The last passage comes from assumption 2.3.

Now, we turn the attention on the inductive step. If $|\alpha| = k$ with $k < d$, there is an index i such that $\alpha_i = 0$. Then

$$\partial^\alpha \mathbf{D}(x) = \int_0^{x_i} \partial^{\alpha + \delta_i} \mathbf{D}(x_{-i}, t) dt,$$

322 where $\alpha + \delta_i$ stands for α with 1 in place of 0 for component i . Therefore,

$$\begin{aligned} \|\partial^\alpha \mathbf{D}\|_{L^2}^2 &= \int_{[0,1]^d} \left(\int_0^{x_i} \partial^{\alpha + \delta_i} \mathbf{D}(x_{-i}, t) dt \right)^2 dx \\ &\leq \int_{[0,1]^d} \int_0^{x_i} (\partial^{\alpha + \delta_i} \mathbf{D}(x_{-i}, t))^2 dt dx \\ &\leq \int_0^1 \int_{[0,1]^d} (\partial^{\alpha + \delta_i} \mathbf{D}(x))^2 dx dt \\ &\leq \int_0^1 4L_2 dt = 4L_2. \end{aligned}$$

323 Here, the first passage is my definition of norm, the second by Jensen's inequality, the measure of
 324 the interval is $x_i \leq 1$, the third one by upper bounding and exchanging the integrals and the last one
 325 comes from induction. $\square$

326 **Theorem 3.2.** Under assumption 2.3, the objective function 2.1 belongs to the space $H_{\text{mix}}^1([0, 1]^d)$
 327 introduced in definition 3.1, and $\|J\|_{H_{\text{mix}}^1}^2 \leq d^2 2^{d+4} L_2$.

Proof. First, by the fact that $\mathbf{p} \geq 0$ component-wise, the objective function reduces to

$$J(\mathbf{p}) = \sum_{i=1}^d p_i \mathbb{P}(\mathbf{B} \leq \mathbf{p} \leq \mathbf{A}) =: \mathbf{D}(\mathbf{p}) \sum_{i=1}^d p_i.$$

Making any partial derivative ∂^α with $\alpha \in \{0, 1\}^d$ gives

$$|\partial^\alpha J(\mathbf{p})| \leq \left| \partial^\alpha \mathbf{D}(\mathbf{p}) \sum_{i=1}^d p_i \right| + |\mathbf{D}(\mathbf{p})| |\alpha| \leq 2d \max_{\alpha \in \{0,1\}^d} |\partial^\alpha \mathbf{D}(\mathbf{p})|,$$

328 the second term being valid since $\partial^\alpha \sum_{i=1}^d p_i = \sum_{i=1}^d \mathbb{I}\{\alpha_i\}$. This proves that

$$\begin{aligned} \|J\|_{H_{\text{mix}}^1}^2 &= \sum_{\alpha \in \{0,1\}^d} \|\partial^\alpha J\|_{L^2}^2 \\ &\leq \sum_{\alpha \in \{0,1\}^d} 4d^2 \max_{\alpha \in \{0,1\}^d} \|\partial^\alpha \mathbf{D}\|_{L^2}^2 \\ &\leq d^2 2^{d+2} \|\partial^\alpha \mathbf{D}\|_{L^2}^2. \end{aligned}$$

329 Lemma B.1 ends the proof. $\square$

Theorem 3.4. [Space equivalence] There is a constant C_d only depending on d such that, for any $f \in H_{\text{mix}}^1([0, 1]^d)$ we have $f \in \mathcal{H}_{\Pi}^k([0, 1]^d)$ and

$$\|f\|_{\mathcal{H}_{\Pi}^k} \leq C_d \|f\|_{H_{\text{mix}}^1}.$$

330 *Proof.* By the Hestenes theorems Vybiral [2006], there is an extension operator $E : H_{\text{mix}}^1([0, 1]^d) \rightarrow$
 331 $H_{\text{mix}}^1(\mathbb{R}^d)$ such that for some constant $C(E)$ independent on f ,

$$\|E(f)\|_{H_{\text{mix}}^1(\mathbb{R}^d)} \leq C(E) \|f\|_{H_{\text{mix}}^1([0,1]^d)}. \quad (4)$$

332 Now, for a function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, let us note $\widehat{g} : \mathbb{R}^d \rightarrow \mathbb{R}$ its Fourier transform. By Plancherel's
 333 theorem, the following identity holds, calling $f_E := E(f)$:

$$\begin{aligned} \|f_E\|_{H_{\text{mix}}^1(\mathbb{R}^d)}^2 &= \sum_{\alpha \in \{0,1\}^d} \|\partial^\alpha f_E\|_{L^2(\mathbb{R}^d)}^2 \\ &= \sum_{\alpha \in \{0,1\}^d} \|\widehat{\partial^\alpha f_E}\|_{L^2(\mathbb{R}^d)}^2 \\ &= \sum_{\alpha \in \{0,1\}^d} \int_{\mathbb{R}^d} \xi^{2\alpha} |\widehat{f_E}(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}^d} |\widehat{f_E}(\xi)|^2 \prod_{i=1}^d (1 + \xi_i^2) d\xi. \end{aligned}$$

334 Here, the first equality follows by definition, the second by Plancherel's theorem, the third one from
 335 the Fourier transform of the derivative and the last one by rearranging the indexes.

Now, it is well known (equation 4.14 [Seeger \[2004\]](#)) that the Fourier Transform of the Matérn kernel k that we defined in eq. (2) is proportional to $(1 + \xi^2)^{-1}$. Therefore, the product kernel k_Π enjoys

$$\widehat{k_\Pi}(\xi) \propto \prod_{i=1}^d (1 + \xi_i^2)^{-1},$$

as the Fourier transform of a multivariate function that factorizes as the product of single-variate function is the product of the Fourier transforms. This shows that, for some constant C_{mat} ,

$$\|f_E\|_{H_{\text{mix}}^1(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} |\widehat{f_E}(\xi)|^2 \prod_{i=1}^d (1 + \xi_i^2) d\xi = C_{\text{mat}} \int_{\mathbb{R}^d} \frac{|\widehat{f_E}(\xi)|^2}{\widehat{k_\Pi}(\xi)} d\xi.$$

By Bochner's theorem [\[Seeger, 2004\]](#), the former writes exactly as the squared-norm of f_E in $\mathcal{H}_\Pi^k(\mathbb{R}^d)$. Therefore, we have proved that,

$$\|f_E\|_{H_{\text{mix}}^1(\mathbb{R}^d)}^2 = C_{\text{mat}} \|f_E\|_{\mathcal{H}_\Pi^k(\mathbb{R}^d)}^2 \geq C_{\text{mat}} \|f\|_{\mathcal{H}_\Pi^k([0,1]^d)}^2,$$

where the last point comes from the fact that f is just f_E restricted to $[0, 1]^d$, by definition. The thesis follows applying (4):

$$\|f\|_{\mathcal{H}_\Pi^k([0,1]^d)} \leq \underbrace{C_{\text{mat}}^{-1} C(E)}_{=: C_d} \|f\|_{H_{\text{mix}}^1([0,1]^d)}.$$

336

□

Theorem 4.2. *The maximum information gain for kernel k_Π is bounded as follows:*

$$\gamma_T \leq C_\gamma \sqrt{T} \log(T)^{d+1/2}.$$

Proof. First, we want to find a bound on the eigenvalues of k_Π . By [Santin and Schaback \[2016\]](#) we have, calling s_i the i -th eigenvalue of k (which is a Matérn kernel of parameters $d = 1, \nu = 1/2$)

$$\mu_i \leq C_1 i^{-2},$$

for some $C_1 > 0$. By construction, the eigenvalue of k_Π are written as a product of the former eigenvalues, for different indices $i_1, \dots, i_j, \dots, i_d$. In particular, the spectrum of k_Π writes as follows:

$$\Lambda(k_\Pi) = \left\{ \prod_{j=1}^d \mu_{i_j} : i_1, \dots, i_j, \dots, i_d \in \mathbb{N} \right\}.$$

337 From this equation, plus the fact that $\mu_i \leq C_1 i^{-2}$, we can upper bound the number of eigenvalues of
 338 k_Π bigger than a threshold ε .

$$|\Lambda(k_\Pi) \cap [\varepsilon, \infty)| \leq \left| \left\{ \prod_{j=1}^d i_j^{-2} \geq \frac{\varepsilon}{C_1} : i_1, \dots, i_j, \dots, i_d \in \mathbb{N} \right\} \right|$$

$$\begin{aligned}
&= \left| \left\{ \prod_{j=1}^d i_j \geq \sqrt{\frac{C_1}{\varepsilon}} : i_1, \dots, i_j, \dots, i_d \in \mathbb{N} \right\} \right| \\
&\leq C_2 \sqrt{\frac{C_1}{\varepsilon}} \log \left(\sqrt{\frac{C_1}{\varepsilon}} \right)^{d-1} \leq C_3 \varepsilon^{-1/2} \log(1/\varepsilon)^{d-1},
\end{aligned}$$

For some constant C_3 , where the last passage comes from equation (2.16) of [Temlyakov and Ullrich \[2016\]](#). Calling $\tilde{\mu}_i$ the i -th eigenvalue of $\mathbf{k}_\Pi$, the former equation shows that

$$\tilde{\mu}_i \leq C_4 i^{-2} \log(i)^{2d-2},$$

339 for some other constant $C_4 > 0$. At this point, Theorem 3 from [Vakili et al. \[2021\]](#)¹ allows to pass
340 from the bound on the eigenvalues to a bound on the information gain as follows:

$$\begin{aligned}
\gamma_T &\leq C_5 \left(\inf_{D \in \mathbb{N}} D \log(T) + T \sum_{i=D}^{\infty} \tilde{\mu}_i \right) \\
&\leq C_6 \left(\inf_{D \in \mathbb{N}} D \log(T) + T \sum_{i=D}^{\infty} i^{-2} \log(i)^{2d-2} \right) \\
&\leq C_7 \left(\inf_{D \in \mathbb{N}} D \log(T) + T \frac{\log(D)^{2d-2}}{D} \right) \\
&= C_8 \sqrt{T} \log(T)^{d+1/2}.
\end{aligned}$$

341 This completes the proof by choosing $C_\gamma = C_8$. □

342 B.1 Proof of the lower bound

Lemma B.2. *Let $F : [0, 1] \rightarrow [0, 1]$ be such that $\int_0^1 |F'(t)| dt \leq 2$ and $F(0) = 0, F(1) = 0$. Then, there are random variables B, A with density functions equal to positive / negative part of F' plus a common constant smaller than one such that*

$$\mathbb{P}(B \leq x \leq A) = F(x).$$

Proof. Call $F'_+(t), F'_-(t)$ the positive/negative parts of $F'(t)$, so that $F'(t) = F'_+(t) - F'_-(t)$ and

$$F_+(x) = \int_0^x F'_+(t) dt \quad F_-(x) = \int_0^x F'_-(t) dt.$$

343 By definition, as $F = F_+ - F_-$ we have

344 1. $F_+(x) \geq F_-(x)$ for all $x \in [0, 1]$, otherwise $F(x)$ would be negative

345 2. $F_+(1) = F_-(1)$

Also, by assumption $\int_0^1 F'_+(t) dt + \int_0^1 F'_-(t) dt \leq 2$, so

$$\int_0^1 F'_+(t) dt = \int_0^1 F'_-(t) dt =: p \leq 1.$$

At this point, we define the two random variables B, A in the following way. Let

$$X \sim \text{Bern}(p) \quad U \sim \text{Unif}(0, 1).$$

If $X = 0$, $A = B = U$. Otherwise,

$$A = F_-^{-1}(pU) \quad B = F_+^{-1}(pU).$$

¹The formula we wrote applies to a kernel with bounded eigenvectors. In our case, this assumption is satisfied as the eigenvectors of $\mathbf{k}_\Pi$ are given by products across different variables of the ones of $\mathbf{k}$, which are bounded

346 Note that, as $F_+ \geq F_-$ point-wise, $A \geq B$ almost surely. Moreover, for any $x \in [0, 1]$

$$\begin{aligned}\mathbb{P}(B \leq x \leq A) &= p \left(\mathbb{P}(F_+^{-1}(pU) \leq x \leq F_-^{-1}(pU)) \right) \\ &= p \left(p^{-1}F_-(x) \leq U \leq p^{-1}F_+(x) \right) \\ &= F_+(x) - F_-(x).\end{aligned}$$

347 Moreover, the density function of B satisfies,

$$\begin{aligned}f_B(x) &= \frac{d}{dx} \mathbb{P}(B \leq x) \\ &= \frac{d}{dx} [(1-p)x] + \frac{d}{dx} [p \mathbb{P}(F_+^{-1}(pU) \leq x)] \\ &= (1-p) + \frac{d}{dx} [p \mathbb{P}(U \leq p^{-1}F_+(x))] \\ &= (1-p) + F'_+(x).\end{aligned}$$

348 with a similar relation holding for A . □

Setup Let us define

$$D(p) := \begin{cases} \frac{4p}{3} & 0 \leq p \leq 1/2 \\ \frac{1}{3p} & 1/2 < p \leq 2/3 \\ \frac{2-2p}{3} & p > 2/3 \end{cases}.$$

349 This function is always bounded between zero and $2/3$. Call, for $K \in 3\mathbb{N}$ and $k \in [2, K-1]$,

$$x_k := \frac{1}{2} + \frac{k}{6K} \quad \phi_k(p) = \left(\frac{\min\{p - x_{k-1}, x_{k+1} - p\}}{\log(\min\{p - x_{k-1}, x_{k+1} - p\})^2} \right)^\beta \mathbb{I}\{x_{k-1} \leq p \leq x_{k+1}\}. \quad (5)$$

350 These functions corresponds to bumps of height $\varepsilon_{K,\beta} := \left(\frac{1}{K \log(K)^2} \right)^\beta$ on the points grid x_k , each
351 with support between the $k-1$ -th and the $k+1$ -th point.

Any algorithm achieves high regret on some instance Consider any algorithm that works on instance $\mathcal{I}_1$ characterized by

$$D_1(p) = D(p) + \phi_1(p).$$

Let T_k the number of times the algorithm pulls $p \in (x_{k-1}, x_{k+1}]$. As $T = \sum_{j=1}^{K/3} \mathbb{E}_{\mathcal{I}_1}[T_{3j+1}]$, there is j such that $\mathbb{E}_{\mathcal{I}_1}[T_{3j+1}] \leq 3T/K$. We fix $k = 3j - 1$, and define an alternative instance as the one with demand function

$$D_2(p) = D(p) + \phi_1(p) + 2\phi_k(p).$$

Let $R_{1,T}$ and $R_{2,T}$ be the regret in the two instances, respectively. By definition, we have

$$R_{T,\mathcal{I}_1} \geq \mathbb{P}_{\mathcal{I}_1}(T_1 \leq T/2) \frac{\varepsilon_{K,\beta} T}{2} \quad R_{T,\mathcal{I}_2} \geq \mathbb{P}_{\mathcal{I}_2}(T_1 > T/2) \frac{\varepsilon_{K,\beta} T}{2}.$$

352 By applying Bretagnolle–Huber inequality (Theorem 14.2 in [Lattimore and Szepesvári \[2020\]](#)) and
353 Lemma 15.1 in [Lattimore and Szepesvári \[2020\]](#),

$$\begin{aligned}R_{T,\mathcal{I}_1} + R_{T,\mathcal{I}_2} &\geq \frac{\varepsilon_{K,\beta} T}{2} (\mathbb{P}_{\mathcal{I}_1}(T_1 \leq T/2) + \mathbb{P}_{\mathcal{I}_2}(T_1 > T/2)) \\ &\geq \frac{\varepsilon_{K,\beta} T}{2} \exp(-\Delta),\end{aligned}$$

where

$$\Delta := \mathbb{E}_{\mathcal{I}_1}[T_j] \left(\text{KL} \left[\text{Ber}(D_1(x_k)) | \text{Ber}(D_2(x_k)) \right] \right).$$

By definition, $D_1(x_k) = \frac{1}{3x_k}$ while $D_2(x_k) = \frac{1}{3x_k} + 2\varepsilon_{K,\beta}$, where $1/2 \leq x_k \leq 2/3$. Therefore,

$$\text{KL} \left[\text{Ber}(D_1(x_k)) | \text{Ber}(D_2(x_k)) \right] \leq -\frac{2}{3} \log(1 + 3\varepsilon_{K,\beta}) - \frac{1}{3} \log(1 - 6\varepsilon_{K,\beta}) \leq 9\varepsilon_{K,\beta}^2.$$

Fixing

$$K = \frac{T^{\frac{1}{1+2\beta}}}{\left(\log T^{\frac{1}{1+2\beta}}\right)^{\frac{4\beta}{1+2\beta}}} \quad \varepsilon_{K,\beta} \approx (T \log^2 T)^{-\frac{\beta}{1+2\beta}},$$

one has $T\varepsilon_{K,\beta}^2 < K$, so that

$$R_{T,\mathcal{I}_1} + R_{T,\mathcal{I}_2} \geq \frac{\varepsilon_{K,\beta} T}{2} \exp\left(-\frac{3T}{K} 9\varepsilon_{K,\beta}^2\right) \geq \frac{\exp(-27)}{2} T\varepsilon_{K,\beta} \gtrsim \frac{T^{\frac{1+\beta}{1+2\beta}}}{\log(T)^{\frac{2\beta}{1+2\beta}}}.$$

The previous instances satisfy assumption 2.3 The previous instances correspond to a demand function that is the sum of

$$D(p) = \begin{cases} \frac{4p}{3} & 0 \leq p \leq 1/2 \\ \frac{1}{3p} & 1/2 < p \leq 2/3, \\ \frac{2-2p}{3} & p > 2/3 \end{cases}$$

whose derivative is

$$D'(p) = \begin{cases} \frac{4}{3} & 0 \leq p \leq 1/2 \\ -\frac{1}{3p^2} & 1/2 < p \leq 2/3, \\ -\frac{2}{3} & p > 2/3 \end{cases}$$

and one (or two) function of the form

$$\phi_k(p) = \left(\frac{\min\{p - x_{k-1}, x_{k+1} - p\}}{\log(\min\{p - x_{k-1}, x_{k+1} - p\})^2} \right)^\beta \mathbb{I}\{x_{k-1} \leq p \leq x_{k+1}\},$$

whose derivative is, for $x_{k-1} \leq p \leq x_k$,

$$\phi'_k(p) = \psi(y) = \beta \frac{y^{\beta-1}(2 - \log(y))}{|\log(y)|^{2\beta+1}} \quad y = p - x_{k-1}$$

and, for $x_k < p \leq x_{k+1}$,

$$\phi'_k(p) = -\psi(y) \quad y = x_{k+1} - p,$$

for ψ defined as before.

We prove that D_1 can be written as $\mathbb{P}(B \leq p \leq A)$ for B, A that satisfy assumption 2.3; the same result is can be proved for D_2 in an analogous way.

D_1 satisfies the assumptions of lemma B.2 for T sufficiently big: $D_1(0) = D_1(1) = 0$ and

$$\begin{aligned} \int_0^1 |D'_1|(x) dx &\leq \int_0^1 |D'(x) + |\phi'_1|(x)| dx \\ &= 2D(1/2) + 2\phi_1(x_1) \\ &= \frac{4}{3} + 2\varepsilon_{K,\beta} \lesssim \frac{4}{3}. \end{aligned}$$

Therefore, lemma B.2 ensures that are random variables B, A with density functions equal to positive / negative part of D'_1 plus a common constant c smaller than one such that

$$\mathbb{P}(B \leq x \leq A) = F(x).$$

What is then to verify that the integral in assumption 2.3 holds for both parts. For the positive one, taking $\alpha = 1/\beta$:

$$\begin{aligned} \int_0^1 D'_1(x)^{1/\beta} dx &= \int_0^1 (c + D'_0(x) + \mathbb{I}\{x \leq x_2\} \psi(x - x_1))^{1/\beta} dp \\ &\leq 3^\beta \int_0^1 c^{1/\beta} + D'_0(x)^{1/\beta} + \mathbb{I}\{x \leq x_2\} \psi(x - x_1)^{1/\beta} dp \\ &\leq 3^\beta \left(1 + (4/3)^{1/\beta} + \int_{x_1}^{x_2} \left(\beta \frac{(x - x_1)^{\beta-1}(2 - \log(x - x_1))}{|\log(x - x_1)|^{2\beta+1}} \right)^{1/\beta} dx \right) \\ &\leq 3^\beta \left(1 + (4/3)^{1/\beta} + \int_0^{1/(6K)} \beta \frac{(x - x_1)^{\beta-1}(2 - \log(x - x_1))}{|\log(x - x_1)|^{2\beta+1}} \right). \end{aligned}$$

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